Project Proposal: Concept-to-Animation — LLM-Generated Narrated Math Explainer Videos

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Problem Statement & Motivation

Animated visual explanations, like those in 3Blue1Brown videos made with the Manim library, are among the most effective ways to teach math, but a single minute of animation takes an expert hours to script and code. We will build a system that takes a short text description of a concept (e.g., "eigenvectors" or "Bayes' theorem") and generates a narrated Manim explainer video with diagrams or graphs. The core research question is which generative techniques make LLM-written animation code render reliably and produce clear, uncluttered visuals, since LLMs frequently hallucinate Manim APIs and place objects that overlap or leave the frame.

Dataset(s)

- **Concept test set (team-curated):** 40 concepts in three difficulty tiers: concrete math (Pythagorean theorem, derivatives); structural (matrix multiplication, binary search, Bayes' theorem); abstract (entropy, recursion, the halting problem).
- **Manim Community documentation and example scenes:** the retrieval corpus for RAG and the source of few-shot examples.

Justification: No standard benchmark scores explanatory animations, so a tiered set lets us measure how performance degrades as concepts get less visual. The official docs ground generation in the current ManimCE API, targeting the most common failure (hallucinated or outdated function calls).

Planned Approach

A multi-stage generation pipeline built in Python with the OpenAI API, LangChain, and an open-source model via Hugging Face:

1. **Planner:** an LLM turns the concept into a storyboard of scenes, each with a visual description and a narration script.
2. **Code generator:** produces ManimCE code for each scene.
3. **Render-and-repair loop:** each scene is rendered; on failure, the traceback is fed back to the LLM for up to N retries.
4. **Layout check:** object bounding boxes are tested for overlap and off-screen placement; defects are sent back for repair.
5. **Narration:** text-to-speech audio synced to animations with the manim-voiceover plugin.

Experiments: an ablation over zero-shot, few-shot, RAG over Manim docs, + error-feedback repair, and + layout repair, run on GPT (API) and an open model (LLaMA-2 or a newer code model, pending approval).

Expected Outputs

1. Results table by technique and model: render success rate, average repair iterations, and layout-defect rate per scene.
2. Human ratings (1–5) of mathematical correctness and clarity on a sample of generated videos, by difficulty tier.
3. Gallery of success and failure cases, including where abstract concepts break down.
4. Demo: type a concept, receive a narrated explainer video.
5. Final report on the best-performing pipeline configuration and its limitations.